

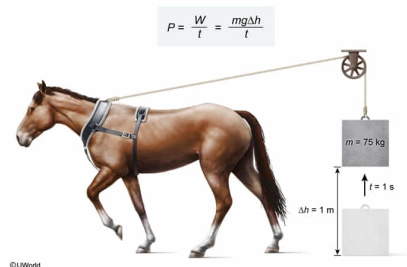
One horsepower (hp) is the amount of power required to lift a 75-kg mass a vertical distance of 1 m in 1 s. What is 2 hp equivalent to in watts (W)?

- ☐ A. 75 W [3%]
- ☐ B. 150 W [36%]
- ☐ C. 750 W [10%]
- ☒ D. 1,500 W [49%]

Correct

48% answered correctly

Explanation:



Mechanical **power** P is the amount of work W done per unit time t :

$$P = \frac{W}{t}$$

From the work-energy theorem, the change in the **potential energy** (PE) of an object is equal to the amount of work done in the vertical direction. Gravitational potential energy is due to an object's position relative to Earth's gravitational field, and it is the product of mass m , gravitational acceleration g , and the height h . Therefore, the **work done** is equal to the change in PE, which depends on the change in height Δh :

$$W = \Delta \text{PE} = m g \Delta h$$

Substituting the above equation into the power equation,

$$P = \frac{m g \Delta h}{t}$$

The SI unit for power is the watt (W), and the **value of g** is 10 m/s^2 . One horsepower (hp) is calculated using the values provided in the question for m , Δh , and t :

$$1 \text{ hp} = 750 \text{ W}$$

Because 1 hp is equal to 750 W, 2 hp is double this value, or **1,500 W**.

(Choices A and B) If the change in gravitational potential energy is calculated without gravitational acceleration g , the incorrectly calculated power of 1 hp is 75 W, and this value doubled is 150 W.

(Choice C) Calculated correctly, 1 hp is equal to 750 W, but the question asks for the value equal to 2 hp.

Educational objective:

Power P is the rate of work W done per unit time t : $P = W/t$. The work done to lift an object is equal to its change in gravitational potential energy ΔPE , which is the product of mass m , gravitational acceleration g , and the change in height Δh : $\Delta \text{PE} = mg\Delta h$.

Time spent: 0 sec
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